When Execution Composition Changes Trace Similarity

A Matched Evaluation of Behavioral Fingerprints

AttackAgent Detect

4 November 2026

Abstract

We study how AI-Agent participation changes errors in linking traces from the same benchmark task. Cyber Task Horizons (CTH) provides Human and Agent traces from capture-the-flag (CTF) challenges. Across 17 CTH tasks, raising the Agent share from 0% to 100% lowers the action-bigram baseline's balanced accuracy (BA) by 10.92 percentage points as missed links rise and false merges fall. The five-share mean BA changes from adding local behavior and sequence features (B/S) are uncertain for all three classifiers. In a frozen external evaluation of 160 NYU D-CIPHER Agent-only challenge groups, B/S raises BA by 5.71 points for XGBoost (95% paired interval [3.63, 7.73]) and 3.64 points for the decision tree ([1.97, 5.37]). Both models miss fewer same-challenge links and falsely merge more different-challenge pairs. For XGBoost, these error rates change by −26.10 and +14.67 points, respectively. SVM loses 8.23 points of BA on the same pairs. On 109 AIT troubleshooting runs, the association gain from observed timing (T) remains uncertain.

1 Introduction

Security investigations must decide which retained traces deserve joint inspection. Event-correlation systems use temporal, causal, provenance, or technique-level evidence to connect records into incident narratives [10, 12, 9, 11]. Many available traces, however, expose only commands, coarse actions, and sometimes timestamps. We use Cyber Task Horizons (CTH) to compare Human and Agent traces within and across capture-the-flag (CTF) challenges.

Figure 1 traces the analysis from observable logs to a windowed fingerprint and two readouts. The pairwise readout uses fingerprints directly as association evidence. The execution-mode readout estimates the recorded Human/Agent label in a separate evaluation.

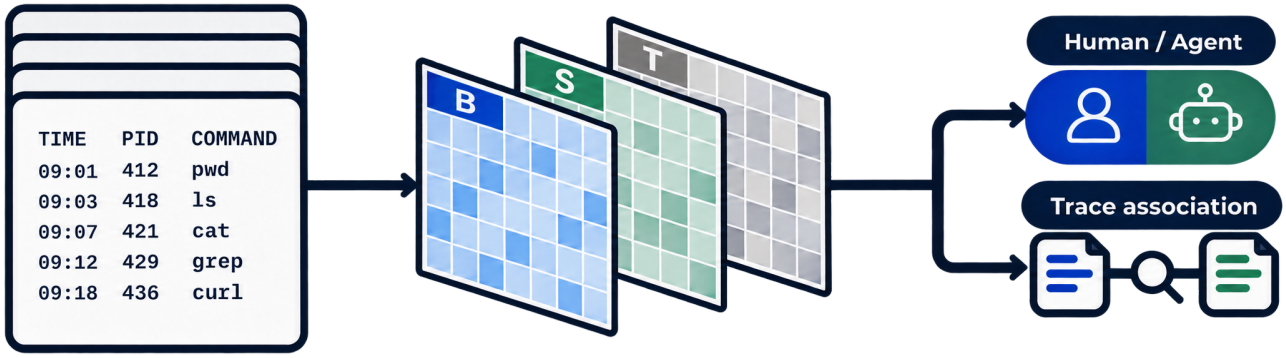


Figure 1: Observable logs, a windowed behavioral fingerprint, and two separately evaluated readouts. The three sheets show the B, S, and T blocks of one trace, joined by columns into one matrix. Each trace is processed independently. Timing is recorded as missing where unobserved. Log fields and cell colors are illustrative.

The pairwise readout has two distinct errors. A *false split* rejects two traces from the same benchmark task. A *false merge* accepts traces from different tasks. We first ask how these errors vary as the share of Agent roles rises from zero to one. We then compare the same models and candidate pairs with and without the fingerprint. The relation label is benchmark task identity, used as a retrieval proxy.

The exact-mixture experiment measures how errors from a fixed similarity threshold change across role allocations. The matched CTH experiment estimates the B/S increment on identical pairs for three learners, and AIT supplies observed launch times for the timing ablation. In the frozen NYU D-CIPHER evaluation, B/S raises BA for XGBoost and the decision tree by recovering more same-challenge links while accepting more different-challenge pairs. SVM loses BA on the same NYU pairs.

Our contributions are: (1) a coupled five-share mixture test that reports false merges and missed links separately; (2) a matched, three-classifier comparison and B/S/T ablation tests across CTH and AIT; and (3) a one-shot, cross-source Agent-only confirmation with explicit error transitions. We state what each task-proxy label permits and retain both unfavorable and uncertain results.

2 Related work

Attack-correlation methods align event, provenance, and technique evidence to reconstruct or search for attack behavior [10, 12, 9, 11, 13]. Agent-trajectory work instead studies workflow representation or identification: Trace analyzes terminal behavior, LIDAR enrolls model–harness fingerprints, Known By Their Actions studies browser-agent traces, and ProcGrep treats coding trajectories as programs [1, 3, 4, 2]. These tasks motivate log-derived behavioral features, but model identity and same-task trace association are different prediction targets. Cross-Agent studies campaign linking with synthetic labels [6]; those labels also differ from the benchmark-task proxy used here. The appendix reports supplementary association results with their source-specific protocols.

3 Representation and association model

Let x and y be two traces, and let $z = 1$ indicate that they share a benchmark task. The base evidence $E(x, y)$ contains command TF–IDF cosine and action-bigram TF–IDF cosine. Each trace also receives a windowed fingerprint: B summarizes local action organization and repetition; S summarizes novelty, revisits, and fragment reuse; T summarizes measured launch gaps or durations only when the source records the required time fields. Missing observations remain missing. No time value is inferred from event order or a workflow label.

Figure 2 defines the eight measurements in each block. With J windows, the trace matrix has shape $J \times 16$ for B/S and $J \times 24$ for B/S/T. The fixed-length readout summarizes each feature column by its mean and standard deviation, with an across-window trend when selected in training. Figure 8 shows measured AIT windows.

For each available block, the pair representation appends the absolute difference between training-standardized trace summaries. In the selected conditional fusion, these differences are multiplied by command similarity before concatenation with E. B/S/T are evidence features for the pair classifier; the task, execution role, source, and path are not model inputs. The classifiers are RBF-SVM, a decision tree, and XGBoost. Their scores are not calibrated probabilities. If either trace lacks a complete required block, the pair uses the same-fold E-only score and threshold; these fallback decisions remain in the evaluation.

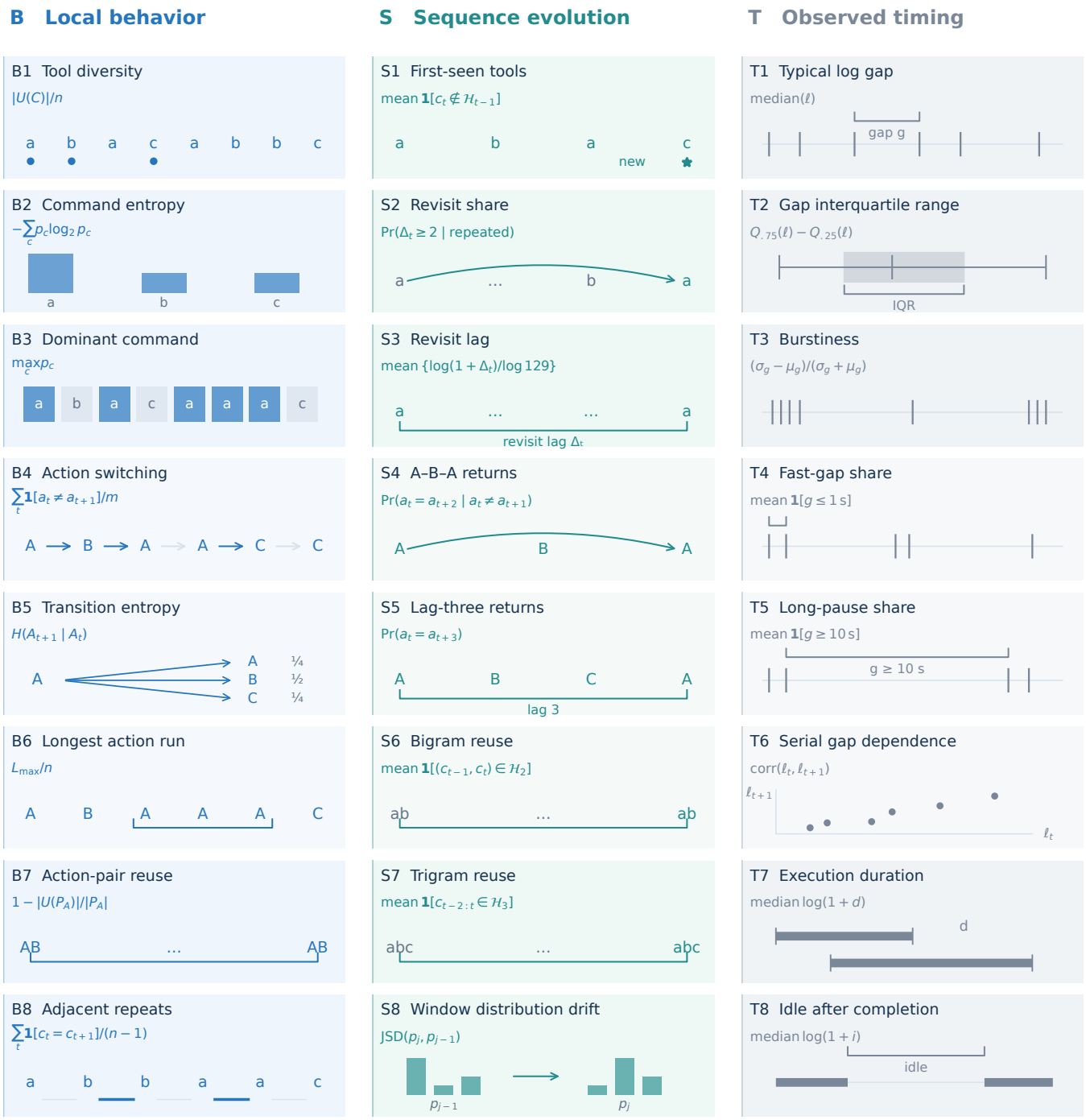


Figure 2: The 24 B/S/T measurements, using the block colors of Figure 1. Glyphs illustrate definitions. Lowercase symbols denote commands and uppercase symbols coarse actions. n counts window events, m valid adjacent action pairs, U unique values, $\mathcal{H}$ prefix history, and Δ_t revisit lag. For observed timing, g is a nonnegative gap in seconds, $\ell = \log(1 + g)$, d duration, and i nonnegative idle time.

4 Evaluation

4.1 Datasets and observed fields

Tables 1 and 2 describe the datasets used in the matched study. CTH supplies the mixed Human/Agent cohort, AIT supplies measured process launch times, and NYU D-CIPHER supplies the external Agent–Agent cohort. Each relation label identifies a benchmark task within its source.

Table 1: Evaluation cohorts. Green robot and blue person icons mark the recorded Agent and Human counts.

Cohort	Records	Mode counts	Task groups	Evaluation role
CTH mixture core	85 raw	51 🤖 + 34 👤	17 CTF tasks	Five Agent-role shares and matched association
AIT audit logs	109 runs	48 🤖 + 61 👤	12 troubleshooting tasks	Timing in same-task association and separate mode recognition
NYU D-CIPHER [15]	585 runs	585 🤖	160 CTF challenges	Frozen external Agent–Agent evaluation

Table 2: Fields available in the trace views used for the matched study. A check marks an observed field and a dash marks an unavailable field.

Source	Trace unit	Order	Start time	Duration	Modes	Fingerprint blocks
CTH mixture core	CTF run	✓	–	–	A 🤖 + H 👤	B/S
AIT audit logs	Task run	✓	✓	–	A 🤖 + H 👤	B/S/T where observed
NYU D-CIPHER	Submission	✓	–	–	A 🤖	B/S

NYU D-CIPHER composition. The evaluation uses four submission snapshots of the same D-CIPHER framework. A trace is an ordered sequence of `run_command` calls from a challenge transcript. Table 3 shows coverage after applying the frozen eligibility rules. The same challenge can appear under multiple snapshots.

Table 3: NYU D-CIPHER submissions. Eligible challenge counts overlap across snapshots. The total counts distinct challenges across all four snapshots.

Model snapshot	Raw transcripts	Eligible traces	Eligible challenges
Claude 3.5 Sonnet	194	147 🤖	147
Claude 3.7 Sonnet	200	128 🤖	128
GPT-4.1	196	153 🤖	153
GPT-4o	197	157 🤖	157
Total	787	585 🤖	160

4.2 Pairing and validation

Exact CTH role mixture. The frozen CTH core has 17 task groups. In 170 coupled allocations, 16 tasks contribute query and candidate roles, with exactly 0, 8, 16, 24, or 32 Agent roles among the 32 allocated roles. Each query has one same-task and three different-task candidates. Candidate tasks and draws are paired across shares. The share is therefore a property of the constructed query galleries, not a resampled proportion of all source records. We retain the earlier command- and action-bigram-cosine baselines to describe this protocol.

Matched classifier and ablation tests. The main CTH comparison uses the same 54,400 candidate-pair decisions for E and E+B/S across all five shares and all three learners. AIT contains 109 independent troubleshooting runs across 12 tasks; each configuration uses the same 43,600 pair decisions, with T computed from recorded launch timestamps. NYU contains 585 runs from four snapshots of one Agent framework, grouped into 160 challenge tasks not used in CTH development. Every NYU query is tested against one same-challenge and three different-challenge candidates. This is an AA-only cross-source test, and a challenge filename is the task-retrieval proxy.

Preprocessing, representation choice, classifier parameters, and thresholds are selected within task-disjoint training folds. Each evidence arm selects its threshold on inner out-of-fold predictions under the FMR-at-most-10% rule; the selected threshold is fixed for the outer evaluation. This constraint is not expected to transfer to the test tasks. CTH results average task outcomes and then the five shares equally; AIT and NYU average task outcomes. We use 10,000 paired task-cluster bootstrap resamples for intervals. Repeated gallery draws and pair decisions within a task are not independent observations, so intervals describe task-cluster uncertainty and remain approximate under shared candidates. No results are adjusted for the many development analyses.

We report balanced accuracy (BA), false-merge rate (FMR), and false-split rate (FSR), where BA $= 1 - (\text{FMR} + \text{FSR})/2$. Increasing one error can therefore be hidden by a reduction in the other; all three measures are shown.

5 Results

5.1 The mixture baseline shifts toward missed links

Figure 3 shows both error rates and BA under the exact CTH role mixture. Command-cosine BA changes from 53.59% at 0% Agent roles to 46.50% at 100% (endpoint difference -7.10 pp, 95% task interval $[-15.45, +0.06]$). Action-bigram BA changes from 57.40% to 46.48% (-10.92 pp, $[-20.61, -3.06]$). At the fixed threshold, missed links increase and false merges decrease from the first endpoint to the last.

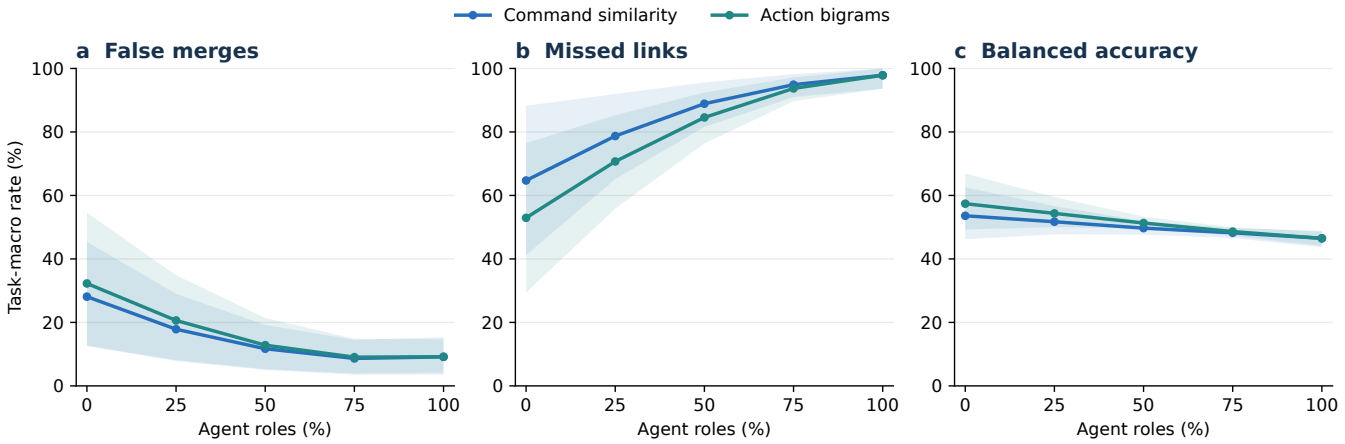


Figure 3: Command and action-bigram cosine baselines at the five exact CTH role shares. From left to right, false merges, missed links, and balanced accuracy. Shading shows 95% intervals from resampling the 17 task groups after averaging paired allocations within each task.

5.2 Matched CTH learners do not show a common B/S gain

Figure 4 shows the same-pair E and E+B/S curves. Averaged equally over the five shares, the XGBoost B/S arm changes BA by -2.38 pp (95% paired interval $[-7.13, +2.25]$), while FMR rises 9.46 points and FSR falls 4.70 points. SVM changes BA by $+1.38$ pp $[-3.83, +6.53]$; its mean FMR also rises. The decision-tree B/S version rejects every candidate, giving 50% BA, zero FMR, and 100% FSR. None of the three learners satisfies the pre-specified requirement of a stable gain without an error trade-off. The small, five-share changes are made explicit in Figure 5; the intervals cross zero.

Table 4: Matched CTH classifier comparison averaged equally over five Agent-role shares. Every classifier uses the same paired candidate decisions. Changes are B/S minus evidence-only E in percentage points; 95% intervals are paired task-cluster bootstrap intervals.

Classifier	BA: E	BA: E+B/S	ΔBA [95% CI]	ΔFMR	ΔFSR
RBF-SVM	52.57	53.95	$+1.38$ $[-3.83, +6.53]$	+6.99	-9.74
Decision tree	52.80	50.00	-2.80 $[-6.61, +0.57]$	-6.59	+12.19
XGBoost	56.06	53.68	-2.38 $[-7.13, +2.25]$	+9.46	-4.70

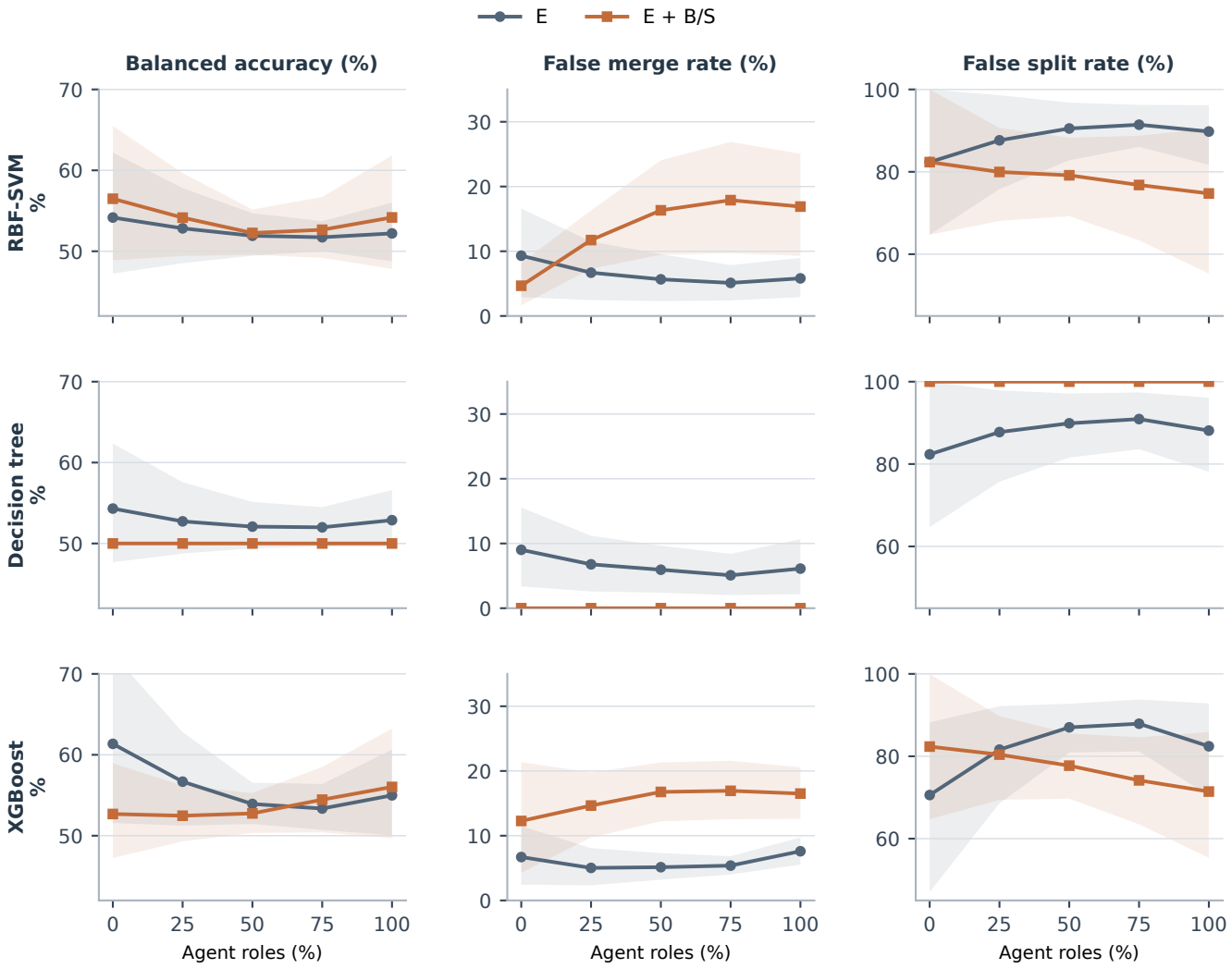


Figure 4: Matched CTH comparison over exact Agent-role shares. Rows are learners; columns are BA, FMR, and FSR. Lines show E and E+B/S, and bands show 95% task-cluster intervals. Y-ranges are shared within each metric column and stated by the axes. The role percentage is over query and candidate roles.

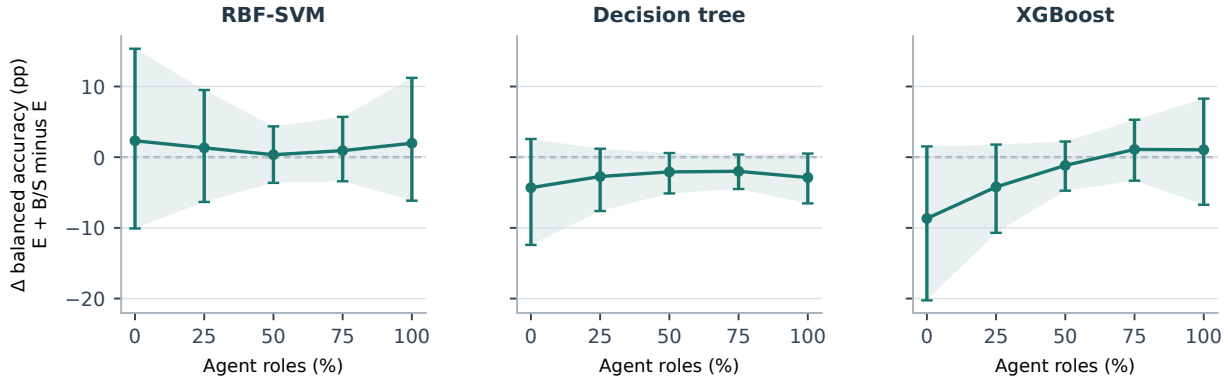


Figure 5: Paired BA change (E+B/S minus E) at each CTH share, in percentage points. The dashed line marks zero; intervals are task-cluster paired 95% intervals.

5.3 Pair type changes the error pattern, not the conclusion

We reuse the same XGBoost predictions and average only the shares where each candidate-pair type exists. For HH, the BA difference is -8.84 pp $[-20.34, +1.67]$; HA is $+1.29$ pp $[-1.78, +4.71]$; AA is $+1.39$ pp $[-5.80, +8.03]$. HA and AA point gains accompany higher FMR, while HH trends downward in BA. No stratum establishes a reliable gain. Pair types describe the modes assigned to the two traces and say nothing about whether their behaviors came from the same real attack or operator (Appendix C).

5.4 AIT timing and external Agent-only transfer

On the AIT association task proxy, adding measured T to E+B/S changes BA by $+0.98$ pp $[-0.21, +2.10]$; the interval spans zero. Removing a single fingerprint block did not isolate a reliable BA benefit in the matched CTH or AIT comparisons. AIT task and collection workflow co-vary with its execution labels, so the timing contrast is limited to this troubleshooting cohort.

On frozen NYU Agent-only challenge retrieval, adding B/S improves BA by $+5.71$ pp $[+3.63, +7.73]$ for XGBoost and $+3.64$ pp $[+1.97, +5.37]$ for the decision tree. With XGBoost, FSR falls from 86.96% to 60.86% and FMR rises from 10.72% to 25.39%. The tree shows the same error trade-off. SVM BA falls by 8.23 pp $[-10.22, -6.28]$. XGBoost also ranks the same-challenge candidate first more often, with a Hit@1 gain of $+3.20$ pp $[+0.45, +5.90]$ (Appendix B).

6 Discussion

On NYU, the reduction in missed links outweighs the increase in false merges for XGBoost and the decision tree under BA's equal weighting of the two error rates. XGBoost's Hit@1 gain supports using its score to prioritize candidates for review on this benchmark. For automatic linking, the observed 25.39% FMR with B/S exceeds the 10% validation ceiling used to select the threshold on CTH. These two uses put different demands on the score and its decision threshold.

The HH/HA/AA breakdown shows why a single overall average is incomplete, but it is descriptive. Its share support and task candidates overlap, and there is no independent mixed-role confirmation dataset. The AIT timing block can be computed from real timestamps, yet its association increment is uncertain and its labels are entangled with task and collection workflow. These results do not identify what caused a particular pair to differ; similar Agent paths, changed execution paths, collection-interface differences, and threshold transfer remain plausible contributors.

All relations are benchmark-task proxies; independent attack-activity labels are absent. CTH terminal and Agent tool-call capture differ, and NYU's snapshots share one framework. Repeated draws do not add independent tasks. Because several alternatives were inspected during development, intervals are exploratory and unadjusted; operational linkage and attribution remain untested.

7 Conclusion

On external Agent-only challenge retrieval, B/S improves balanced accuracy for XGBoost and the decision tree, with fewer missed links and more false merges. XGBoost also improves the ranking of the correct candidate. SVM loses BA on the same pairs, and the five-share CTH experiment and AIT timing ablation leave their respective BA increments uncertain. For pair decisions, reporting both error rates alongside BA makes the cost of recovering more links visible.

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A Supplementary CTH and external results

Figure 6 displays paired changes by role type, while Table 7 gives exact XGBoost values by mixture share. The external table reports learner-specific absolute results and the paired error changes; its intervals cluster the 160 held-out challenge groups.

Table 5: Frozen NYU D-CIPHER AA challenge-retrieval comparison. The challenge filename defines the same-task proxy. Changes are E+B/S minus E; intervals cluster by 160 task groups.

Classifier	BA: E	BA: E+B/S	ΔBA [95% CI]	FMR: E+B/S	ΔFMR	ΔFSR
RBF-SVM	58.16	49.93	−8.23 [−10.22, −6.28]	0.13	-26.45	+42.91
Decision tree	49.69	53.33	+3.64 [+1.97, +5.37]	8.54	+4.69	-11.98
XGBoost	51.16	56.87	+5.71 [+3.63, +7.73]	25.39	+14.67	-26.10

Table 6: CTH XGBoost error changes by pair type, averaged equally over the role shares where the type exists. HH, HA, and AA are trace-role pairs, not activity or actor labels.

Pair type	ΔBA [95% CI]	ΔFMR	ΔFSR
Human–Human	−8.84 [−20.34, +1.67]	+5.92	+11.76
Human–Agent	+1.29 [−1.78, +4.71]	+15.66	-18.23
Agent–Agent	+1.39 [−5.80, +8.03]	+9.77	-12.54

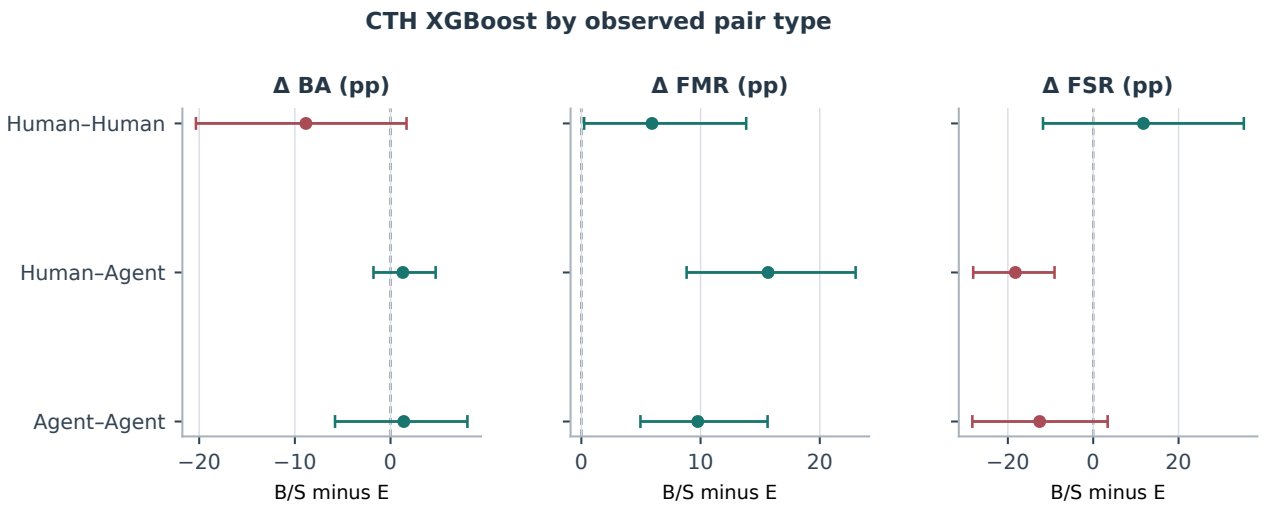


Figure 6: CTH XGBoost change (E+B/S minus E) by pair type and metric. Points are share-equal means over the eligible proportions; bars show paired task-cluster 95% intervals. HH, HA, and AA are evaluation roles, not attack identities.

Table 7: CTH XGBoost results by exact Agent-role share. Values are task-macro means over the same paired candidates; ΔBA intervals preserve task clustering.

Agent roles	BA: E	BA: E+B/S	ΔBA [95% CI]	FMR: E+B/S	FSR: E+B/S
0%	61.35	52.68	−8.67 [−20.26, +1.53]	12.28	82.35
25%	56.68	52.47	−4.21 [−10.70, +1.79]	14.66	80.40
50%	53.92	52.76	−1.16 [−4.74, +2.22]	16.76	77.72
75%	53.36	54.45	+1.10 [−3.33, +5.29]	16.94	74.15
100%	54.99	56.03	+1.04 [−6.73, +8.28]	16.51	71.43

B Frozen external evaluation protocol

The NYU task manifest was frozen before fitting. Four public D-CIPHER snapshots from the official NYU-LLM-CTF leaderboard-submissions repository were included [15]. The exact source revision and hashes are in the frozen evaluation manifest. Challenge groups overlapping the CTH development pool and one duplicate command sequence were removed. The remaining 160 groups and 585 runs were partitioned by a fixed hash into three retrieval galleries. For each run query, 100 deterministic draws selected one other run from the same challenge and one run from each of three different challenges in the same partition. All three classifiers and both feature arms used the identical 234,000-pair list. Training used only the 17 CTH development task groups and all available training pairs; model parameters and thresholds were not selected from NYU outcomes. The CTH-derived threshold did not preserve its target FMR on the external source.

The official repository and snapshot metadata identify challenge filenames and run configuration. A shared challenge name is only the relation proxy. The four snapshots all belong to one Agent framework; this dataset cannot test Human–Agent associations or activity/actor attribution. Error transitions are counts of repeated pair decisions, not independent events: XGBoost corrected 11,241 earlier false merges and introduced 37,348 new false merges, while it corrected 19,708 earlier false splits and introduced 4,329 new false splits. These counts explain the rate changes but do not increase the independent sample size.

C Pair-type support and interpretation

The CTH type-specific estimates reuse the outer predictions. HH is supported at 0–75% Agent roles, HA at 25–75%, and AA at 25–100%. Each type is averaged equally across the shares where it exists and then across tasks. At 50%, the XGBoost gallery contributes 662 positive and 2,004 negative HH pair decisions, 1,396 positive and 4,161 negative HA decisions, and 662 positive and 1,995 negative AA decisions. Pair decisions share queries, candidate tasks, and task clusters; the counts are denominators, not independent sample sizes. The interval widths and overlapping task support preclude interpreting the three rows as independent replications.

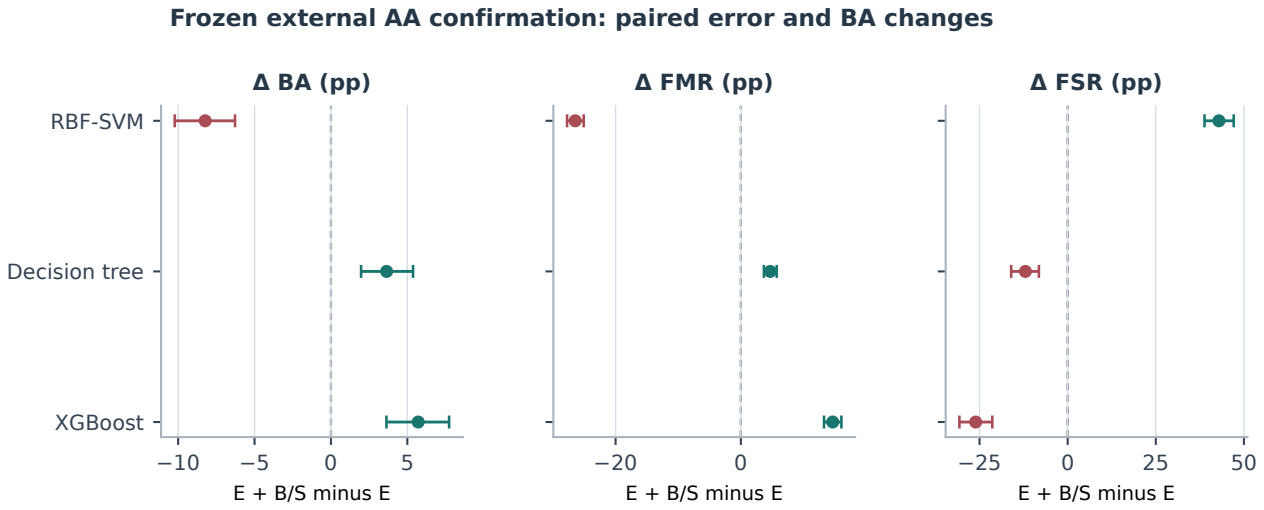


Figure 7: Frozen NYU AA evaluation: paired changes in BA, FMR, and FSR for E+B/S relative to E. The vertical dashed reference is zero; intervals cluster 160 challenge groups.

D Fingerprint definitions and observed AIT traces

Table 8 lists the feature definitions in Figure 2. The AIT examples connect the timing block to measured launch sequences and a separate execution-mode recognition experiment.

Table 8: Window-level feature definitions. A value is missing when its required observations are absent or insufficient.

Block	ID	Measurement
B	B1–B3	Distinct command share; Shannon entropy of command names; share of the dominant command.
B	B4–B6	Fraction of known adjacent action pairs that switch group; conditional entropy of the next action; longest single-action run divided by window length.
B	B7–B8	One minus the unique mapped adjacent-action-pair share; fraction of adjacent command names that are identical.
S	S1–S3	First-seen command share; fraction of repeated commands whose previous occurrence is at least two positions back; mean normalized log revisit span.
S	S4–S5	A-B-A action return share among valid triplets; share of valid four-action segments whose first and fourth action match.
S	S6–S8	Reused adjacent command-bigram share; reused command-trigram share; Jensen–Shannon divergence between current and preceding window action distributions.
T	T1–T3	Median log(1+gap); interquartile range of log gaps; burstiness $(\sigma - \mu)/(\sigma + \mu)$ over nonnegative raw gaps.
T	T4–T6	Fraction of gaps at most one second; fraction at least ten seconds; lag-one correlation of log gaps when at least three valid gaps exist.
T	T7–T8	Median log(1+observed event duration); median log(1+idle interval) after subtracting observed duration from the next start time.

D.1 Observed AIT behavior and timing



Figure 8: Two AIT traces from the same troubleshooting task, visualized as 24 B/S/T columns over variable-length windows. The shared blue–amber scale shows cohort-standardized values for display. Hatched cells mark missing observations.

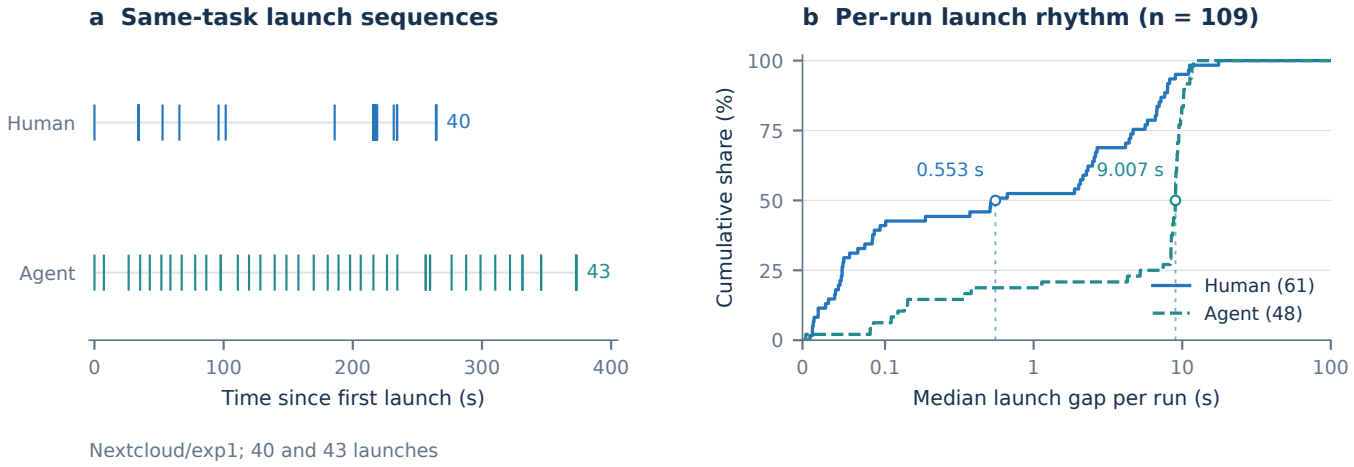


Figure 9: Observed timing behind T. Left, real launch timestamps from the two traces in Figure 8, aligned to their own first observed launch with a shared seconds scale. Right, empirical distributions of one median launch gap per run over all 109 AIT runs. The scale is logarithmic above 0.1 seconds and preserves zero gaps. Each run contributes at most its first 128 login-associated events.

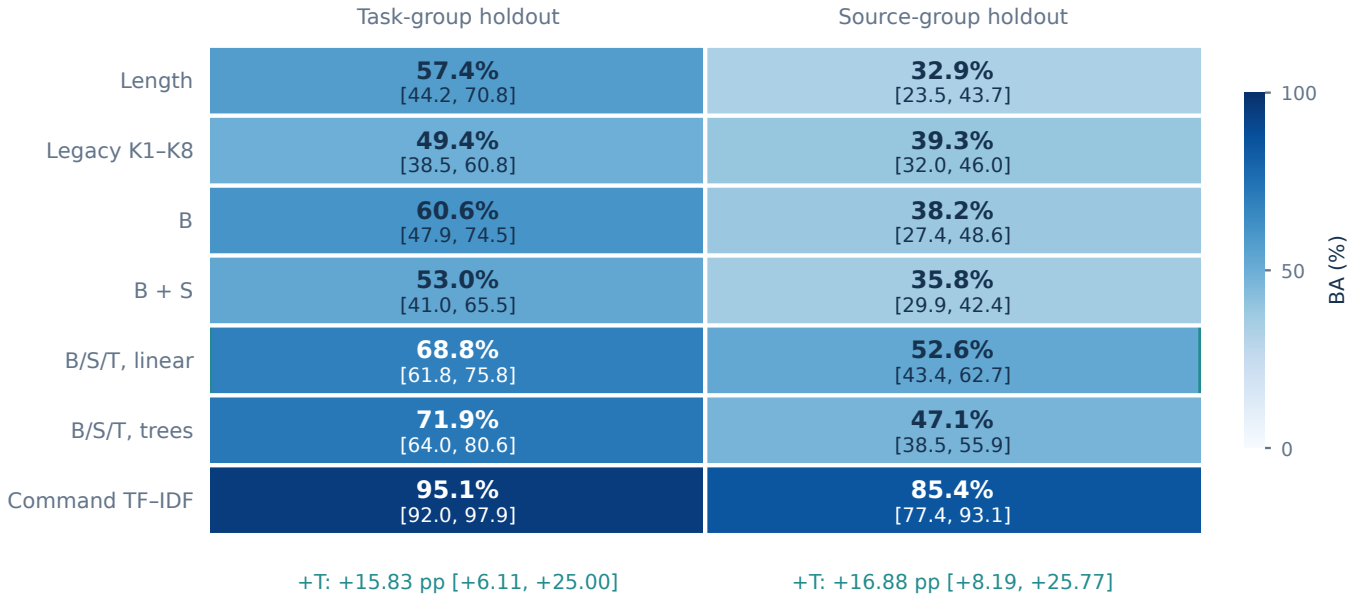


Figure 10: Earlier execution-mode recognition on 109 AIT runs (61 Human, 48 Agent in 12 task groups). Cells report balanced accuracy and 95% intervals for the recorded mode label. The outlined row shows the linear B/S/T readout and the bottom annotations show its paired T increment. Collection workflow co-varies with the released labels.

E Conditioned association across sources

The within-source comparisons in Figure 12 and the source-held-out comparisons in Table 9 use the conditioned shallow-tree protocol. The frozen NYU test uses the matched learners and the CTH-trained thresholds described in Appendix B.

Table 9: Complete source-held-out association comparison. Training uses the other sources; B/S windows contain eight events. BA is in percent; differences and unadjusted 95% cluster intervals are in percentage points. Green/red encode improvement/decline in the point estimate; interval bounds retain the uncertainty.

Test source	Evidence	+ B/S	ΔBA [95% CI]
CTH	51.98	56.69	+4.71 [+0.69, +8.83]
KYPO	66.40	66.61	+0.21 [−1.06, +1.36]
Antieval	71.90	68.63	−3.27 [−6.28, −0.22]

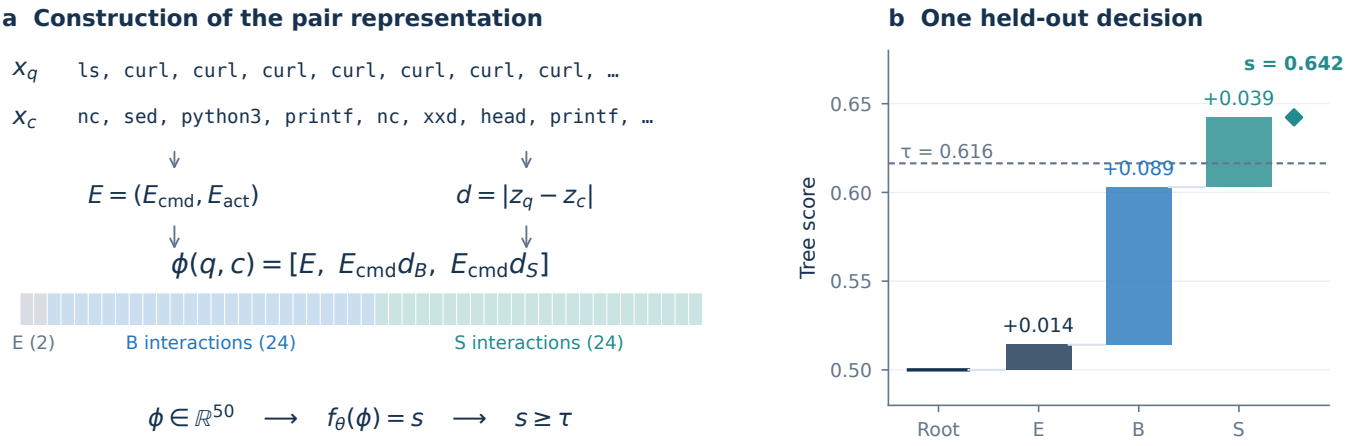


Figure 11: Earlier conditioned tree scorer and one held-out CTH `locktalk` decision. Panel a shows how two evidence scores and 48 conditioned B/S differences enter the scorer. Panel b shows contributions to a selected score among 31 corrected cross-mode same-task pairs in the purged source holdout. Each model uses its own training-selected threshold.

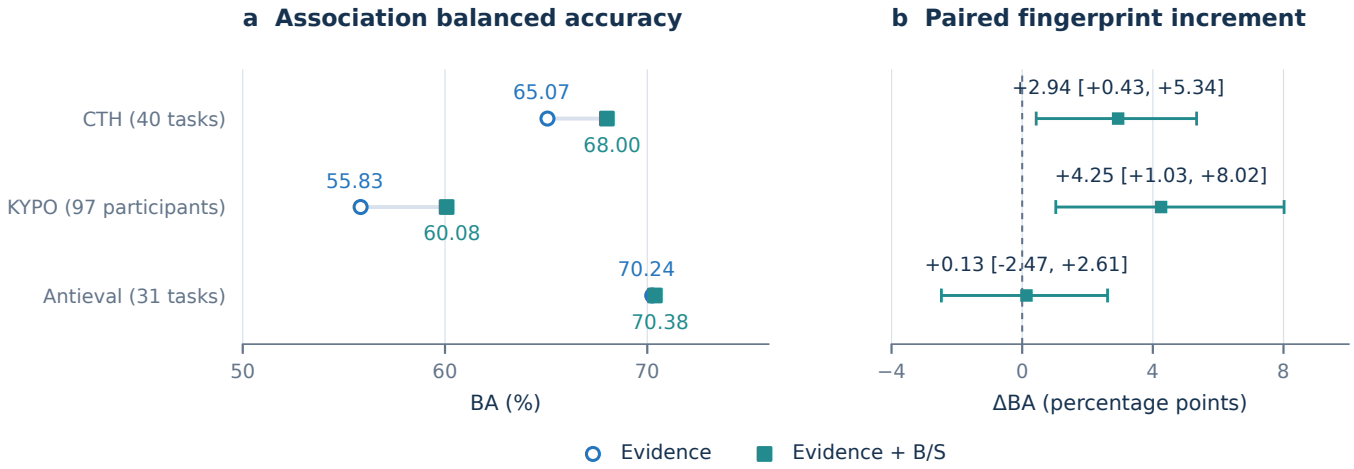


Figure 12: Earlier within-source evidence-conditioned association. Panel a compares the same shallow-tree family before and after adding B/S. Panel b shows paired BA changes and 95% intervals. CTH and Antieval resample tasks. KYPO resamples query participants with galleries fixed. The intervals are exploratory and unadjusted.